STUDENT PERFORMANCE ANALYSIS SYSTEM

Project Overview & Viva Reference

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1. Project objective

The application centralises student profiles, subject records and semester marks. It calculates grades and pass/fail status, visualises academic trends, highlights learning gaps and exports class or individual-ready datasets.

2. Technology stack

Python 3 · Streamlit · Pandas · NumPy · SQLite · Plotly · ReportLab · OpenPyXL

3. Main modules

Module	Responsibility
app.py	Streamlit interface, forms, navigation and downloads
database.py	SQLite schema, CRUD operations and demo data
analysis.py	Grades, class summary, ranking, insights and trend estimate
visualization.py	Interactive Plotly charts
report_generator.py	PDF and Excel report generation

4. Academic calculations

Total = Internal + External. Percentage = Total ÷ Maximum Marks × 100. A student passes at 40% or above. Grade boundaries are centralised in analysis.py and can be customised.

5. Viva talking points

• Streamlit is used to build a Python web interface without a separate frontend framework.
• SQLite stores relational data in three linked tables.
• Pandas performs grouping and aggregation for class and subject analysis.
• Plotly produces interactive visualisations.
• A least-squares line estimates a next-semester score when multiple semester observations exist; it is a basic educational estimate, not a guarantee.

6. Run locally on Windows

```
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
streamlit run app.py
```